

Bayesian Interpretation of BLAST Evidence

(Basic Local Alignment Search Tool)

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Outline

1. Introduction to BLAST
2. BLAST Algorithm Details
3. Bayesian Interpretation
4. Case study: Hemoglobin
5. Sequential Updating
6. Case Study: Species Gene Matching



Introduction to BLAST



Local Alignment

Query

ATGCTACGTTACCGTAACTG

Database sequence

GGATCCCGTTACCGTAAGGA

Main idea

BLAST compares a query against a database and looks for strong local alignments.

Why locality matters

Two sequences can share one conserved region even when they are not similar end to end.



Biological Hypotheses

Why compare sequences?

Similar DNA or protein sequences often indicate shared function, shared structure, or shared evolutionary origin. Sequence matching lets us use known biological information to interpret an unknown query.

What BLAST is used for

- identifying candidate homologs
- annotating newly sequenced genes or proteins
- locating related regions in genomes
- comparing species and tracing evolutionary relationships
- checking whether a query belongs to a known family

DNA matches can suggest

gene identity, conserved genomic regions, regulatory context, or locus-level relationships

Protein matches can suggest

conserved function, structural similarity, functional domains, and stronger evolutionary signal across distant species

Why BLAST became standard

It is fast, database-based, and useful long before a full probabilistic model is built.



Types of BLAST

Different BLAST programs are used depending on whether the query and database are nucleotide sequences, protein sequences, or one must be translated into amino acids before comparison.

Program	Query	Database	Typical use
<i>blastn</i>	DNA/RNA nucleotide sequence	nucleotide database	Find similar genes, genomic regions, or loci
<i>blastp</i>	protein sequence	protein database	Compare functional proteins and detect conserved protein families
<i>blastx</i>	nucleotide sequence translated in all reading frames	protein database	Search a DNA query when the biologically meaningful signal is protein-level conservation
<i>tblastn</i>	protein sequence	nucleotide database translated in all reading frames	Search a protein against genomic or transcript nucleotide data

Main distinction: protein-based searches are often more informative across distant species because amino acid sequences are usually more conserved than raw DNA sequence.



BLAST Algorithm Details



Challenges for Sequence Matching

- **Substitutions:** related sequences may differ at individual positions while preserving similar function.
- **Indels:** insertions and deletions shift the alignment and make direct position-by-position comparison unreliable.
- **Reading frame:** DNA is interpreted in triplet codons, so frame shifts can obscure protein-level similarity, ESPECIALLY in translation searches.

ATGCTACGTTACCGTAACTG

ATGCTACATTACCGTAACTG

ATGCTACGTTACCGTAACTG

ATGCTAC--TACCGTAACTG

ATG | GAA | CTT | GCC

TGG | AAC | TTG | CCA

BLAST must score similarity flexibly enough to recover true relatedness even when exact sequence identity has been disrupted.



BLAST Scoring

Matches and substitutions

In a local alignment, each aligned position contributes to the total score.

For DNA, a simple scheme is

match > 0 , mismatch < 0 .

Example:

```
A T G C T A
A T A C T A
```

Here most positions agree, while one substitution lowers the score.

The total score combines rewards for matches with penalties for mismatches and gaps.

Gaps

Insertions and deletions are represented by gaps in the alignment.

Example:

```
A T G C T A C G
A T G - - A C G
```

The usual rule is an affine penalty:

$$a + b(k - 1)$$

Opening a new gap costs more than extending an existing one



Protein Substitution Matrices

In protein alignment, not all substitutions should be treated equally.

- Some substitutions are **conservative**: replacing one amino acid with another of similar chemical character may preserve structure or function better than a drastic replacement.
- BLAST therefore does not score every protein mismatch the same way. Instead, amino-acid pairs are scored using a **substitution matrix**.
- Two major matrix families are **PAM** and **BLOSUM**.
- Later quantities such as raw score, bit score, and E-value depend on the scoring system used for the alignment.



Bit Scores and E-values

Bit score

The raw alignment score S depends on the scoring system, so by itself it is not directly comparable across different substitution matrices or gap penalties.

BLAST standardizes the raw score using the Karlin-Altschul parameters K and λ :

$$S' = \frac{\lambda S - \ln K}{\ln 2}$$

The quantity S' is the **bit score**. It is a normalized version of the raw score that is easier to compare across searches.

E-value

Once the raw score is calibrated, BLAST estimates how surprising that score would be under a null model of random sequence similarity.

The E-value can be written as

$$E \approx Kmne^{-\lambda S}$$

or equivalently in terms of the bit score as

$$E \approx mn2^{-S'}$$

Here m and n denote the effective query and database lengths. The E-value is the expected number of chance hits this good or better in the search (similar to a p value). 0 - 10



Bayesian Interpretation



Bayesian Inference

Instead of asking whether a hit is surprising under a null model, we ask for the probability that it represents a true homolog after observing the BLAST evidence.

$$P(H = 1 \mid y) = \frac{P(H = 1) f_1(y)}{P(H = 1) f_1(y) + P(H = 0) f_0(y)}$$

H	$P(H = 1), P(H = 0)$	$f_1(y), f_0(y)$	y
indicator of class	prior probabilities	class-conditional	transformed observed
$H = 1$: true homolog	prevalence of homologs	distribution of the	evidence
$H = 0$: non-homolog	before seeing the hit	observed	(negative log E-value)
		BLAST evidence under	
		each class	

The output is a posterior probability between 0 and 1, which is different in meaning from an E-value.



Example Calculation

Suppose the observed BLAST evidence is $Y = 6$, corresponding for example to an E-value near 10^{-6} after transformation.

Assume

$$P(H = 1) = 0.2, \quad P(H = 0) = 0.8$$

and

$$f_1(6) = 0.30, \quad f_0(6) = 0.02.$$

Then

$$P(H = 1 \mid Y = 6) = \frac{P(H = 1)f_1(6)}{P(H = 1)f_1(6) + P(H = 0)f_0(6)} = \frac{0.2 \cdot 0.30}{0.2 \cdot 0.30 + 0.8 \cdot 0.02} \approx 0.79.$$

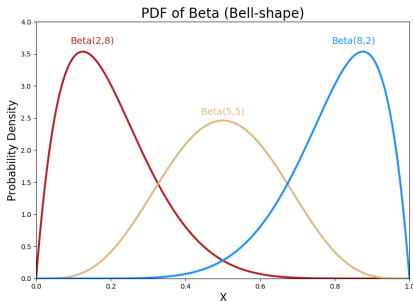
Even with a prior probability of only 0.2, strong observed evidence can raise the posterior probability of true homology to about 0.79.



Uncertain Priors

Instead of fixing the homolog probability at one value, we can treat it as an unknown parameter $\pi = P(H = 1)$ and place a prior distribution on it.

$$\pi \sim \text{Beta}(a, b)$$



- π is still a probability, so it must lie between 0 and 1.
- The Beta distribution describes our uncertainty about which value of π is most plausible before seeing the data.
- Different choices of a and b encode different prior beliefs about how common true homologs are.

The transformed BLAST evidence is

$$Y = -\log_{10}(E^*), \quad E^* = \max(E, 10^{-300}).$$

We model the observed evidence conditionally on the same parameter $\pi \in [0, 1]$ used in the prior:

$$Y \mid \pi \sim \mathcal{N}(100\pi, \sigma^2).$$

- The parameter π scales the expected strength of the BLAST evidence.
- If $\pi = 1$, then the mean of Y is 100, corresponding to an E-value near 10^{-100} .
- If $\pi = 0$, then the mean of Y is 0, corresponding to an E-value near 1.

Metropolis-Hastings Sampling

The posterior is specified only up to proportionality:

$$p(\theta \mid Y) \propto p(\theta) f_1(Y \mid \theta).$$

- We use **Metropolis-Hastings** because it does not require the normalizing constant. The acceptance step depends only on ratios of the posterior, so the unknown denominator cancels.
- This makes it well suited to our model, where the main emphasis is on the homolog evidence term f_1 rather than a fully normalized closed-form joint posterior.
- Once posterior samples are available, **all posterior summaries of interest can be estimated from the sample**, including MAP, posterior means, variances, and credible intervals.



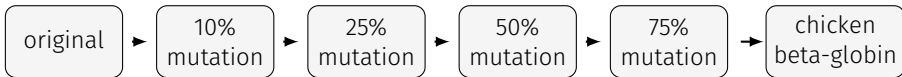
Case study: Hemoglobin



BLAST Evidence for Hemoglobin Subunit Beta

Hemoglobin subunit beta is one of the protein components of hemoglobin, the oxygen-transport protein in red blood cells. It is 147 Amino Acids long

All query proteins were taken from the UniProt protein database and compared against the original hemoglobin subunit beta sequence as the fixed target.



This sequence of queries creates a controlled similarity ladder, ranging from the original protein to heavily mutated versions and then to a real homolog from another species.

The goal is to see how BLAST scores, E-values, and the Bayesian interpretation change as the query moves farther from the fixed reference.

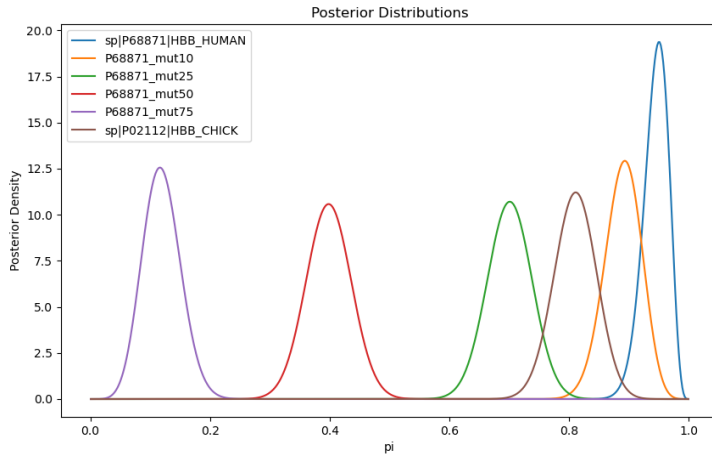


Posterior Summaries

Query	E-value	Mean	Std	$q_{0.025}$	$q_{0.975}$
Original human	4.0×10^{-107}	0.9438	0.0211	0.8958	0.9784
10% mutant	6.0×10^{-95}	0.8889	0.0307	0.8260	0.9460
25% mutant	6.0×10^{-72}	0.6996	0.0380	0.6254	0.7758
50% mutant	5.0×10^{-40}	0.3979	0.0386	0.3204	0.4744
75% mutant	7.0×10^{-8}	0.1205	0.0319	0.0600	0.1857
Chicken beta-globin	3.0×10^{-84}	0.8093	0.0351	0.7377	0.8775



Posterior Trend



Sequential Updating



Sequential Updating

A long genomic DNA sequence can be split into fragments, and each fragment can contribute a new Bayesian update about whether the source contains a target gene or locus.

$$P(G = 1 \mid D_1) \rightarrow P(G = 1 \mid D_1, D_2) \rightarrow \cdots \rightarrow P(G = 1 \mid D_1, \dots, D_k)$$

- Here $G = 1$ means the source contains the target gene or locus, and D_i denotes the BLAST evidence from fragment i .
- This is a natural extension of the Bayesian framework because uncertainty is updated as information accumulates.

The main question is no longer whether one hit looks convincing by itself, but how confidence changes as multiple fragments are observed.



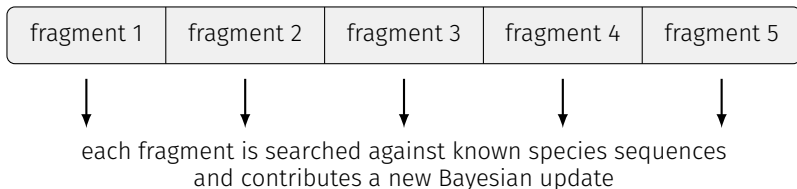
Case Study: Species Gene Matching



DNA Fragment Updates

The DNA example is based on the human beta-globin locus: *Homo sapiens* hemoglobin subunit beta (*HBB*), RefSeqGene (LRG_1232) on chromosome 11, with total length 10106 bp.

BLAST is not well suited to using the entire long genomic sequence at once, so we instead divide the locus into shorter fragments (200 bp) and analyze them sequentially.



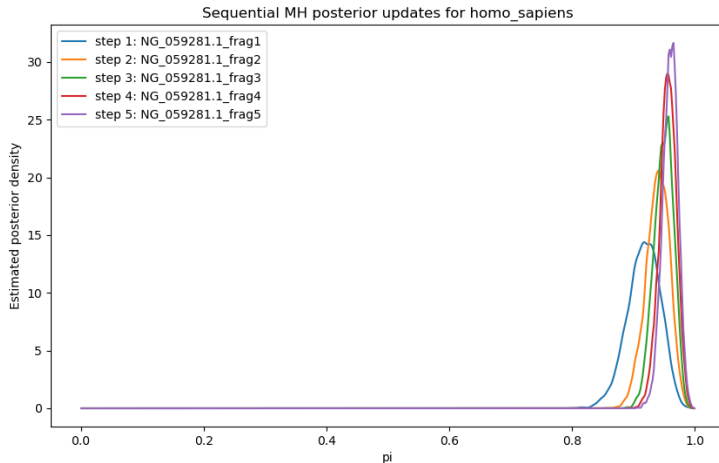
For each fragment, we take the top BLAST hits from 3 known species in NIH databases and update the posterior probability that the source gene belongs to a given species.

Final Posterior Summaries

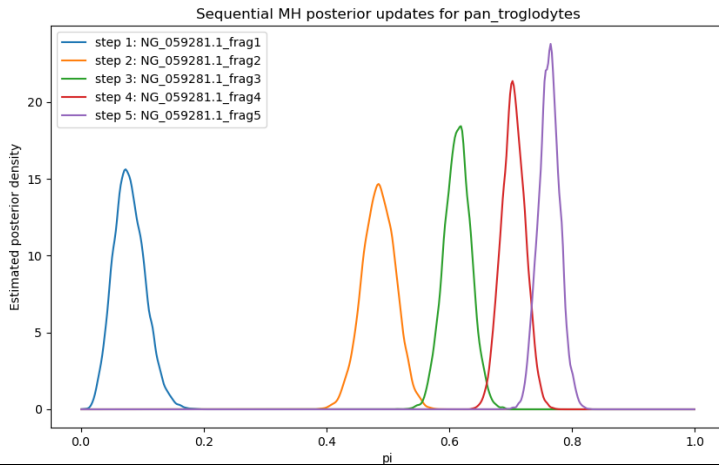
Species	E_1	E_2	E_3	E_4	E_5	Mean	Std	$q_{0.025}$	$q_{0.975}$
Human	7×10^{-100}	7×10^{-100}	7×10^{-100}	7×10^{-100}	7×10^{-100}	0.9596	0.0125	0.9337	0.9823
Chimp	1	1×10^{-97}	4×10^{-88}	3×10^{-99}	7×10^{-101}	0.7625	0.0174	0.7287	0.7976
Mouse	1	1	1	5×10^{-60}	3×10^{-42}	0.2058	0.0169	0.1732	0.2395



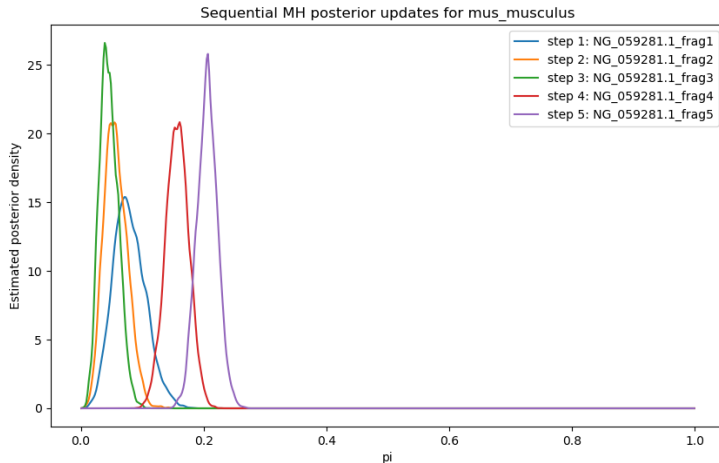
Human Posterior Evolution



Chimp Posterior Evolution



Mouse Posterior Evolution



Next steps

- **Prior sensitivity analysis:** test how strongly the posterior depends on the Beta prior parameters and other prior choices.
- **Translated search:** extend the framework to translated DNA–protein comparisons, where frameshifts and alternative splicing can complicate interpretation.
- **Hierarchical models for BLAST settings:** place additional priors on scoring parameters so statistical uncertainty is accounted for.
- **Different likelihoods:** compare the current normal likelihood with alternative models that may better capture skewness, heavy tails, or other features of BLAST-derived evidence.



THANK YOU

QUESTIONS?

